

On the Real and Imaginary Parts of Kronecker-Sparse Factorizations, with Application to Fast Transforms

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Abstract—Fast transforms such as Discrete Cosine, Sine and Hartley Transforms can be expressed as the real or the imaginary part of a product of complex sparse matrices. While products of Kronecker-sparse factors, a general class of structured sparse matrices encompassing square dyadic butterfly and monarch matrices, are known to represent many fast transforms efficiently, it has remained unclear whether taking the real or the imaginary part of such a product preserves this structure. In this letter, we give a necessary and sufficient condition on the factors to ensure that the structure is preserved. This condition is satisfied in the case of the Discrete Cosine Transform, the Discrete Sine Transform and the Discrete Hartley Transform, allowing us to provide a new representation of these transforms as products of Kronecker-sparse factors.

Index Terms—Fast transforms, Discrete Cosine Transform, sparse factorization, structured matrices.

I. INTRODUCTION

FAST transforms are a fundamental tool in signal processing and machine learning, enabling efficient computation of linear operations on structured data. In many applications, it is desirable to approximate a given linear operator by a product of sparse factors, which can be efficiently implemented and stored. Such factorizations reduce the computational and storage costs of applying the operator from quadratic to near-linear in the matrix dimension, e.g., the Fast Fourier transform [1] and the Fast Wavelet transform [2], which is critical for large-scale signal processing and deep learning pipelines.

Recently, a new general class of sparse matrices, called Kronecker-sparse factors, has been introduced in [3]. These factors arise in the context of factorizing dense matrices and many fast transforms can be expressed as products of Kronecker-sparse factors, enabling efficient computational and storage costs.

Among the various class of fast transforms, a part of them can be expressed as the real or the imaginary part of a product of complex Kronecker-sparse factors, such as the Discrete Cosine Transform (DCT) [4], the Discrete Sine Transform (DST) and the Discrete Hartley Transform (DHT) [5]. However, it is

not clear whether the real or the imaginary part of a Kronecker-sparse factorization is still a Kronecker-sparse factorization. To the best of our knowledge, this question has not been addressed in the literature, therefore, the contribution of this letter is to give a necessary and sufficient condition on the factors to ensure that the real or the imaginary part of a product of Kronecker-sparse factors is still a product of Kronecker-sparse factors.

In the following, we denote by $\text{Re}(\cdot)$ and $\text{Im}(\cdot)$ the real and imaginary part of a complex number, respectively. Vectors (resp. matrices) are denoted by lowercase (resp. uppercase) bold letters. $\mathbf{1}_{m \times n}$ denotes the matrix of size $m \times n$ with all entries equal to 1. $\mathbf{I}_n$ denotes the identity matrix of size $n \times n$. Finally, for any $n \in \mathbb{N}$, we denote by $\llbracket 1, n \rrbracket$ the set of integers $\{1, \dots, n\}$.

The letter is organized as follows: in Section II, we introduce the Kronecker-sparse factors; we derive, in Section III, the necessary and sufficient condition; finally, we conclude in Section IV.

II. DEFINITION OF THE SPARSITY PATTERN

We begin by introducing our class of structured sparse matrices, called Kronecker-sparse factors, which are defined by their support.

Definition 1 (Kronecker-sparse support and their chain [3, Definition 4.1]): For $a, b, c, d \in \mathbb{N}$, a Kronecker-sparse factor of chain $\pi := (a, b, c, d)$ is a matrix in $\mathbb{R}^{abd \times acd}$ or $\mathbb{C}^{abd \times acd}$, such that its support is included in $\mathbf{S}_\pi := \mathbf{I}_a \otimes \mathbf{1}_{b \times c} \otimes \mathbf{I}_d \in \{0, 1\}^{abd \times acd}$. The set of Kronecker-sparse factors associates with the chain π is denoted by $\mathcal{K}_\pi$.

According to [3, Lemma 4.3], for a Kronecker-sparse factor of chain $\pi = (a, b, c, d)$, the number of non-zero entries is $abcd$. Therefore, the factor is sparse if $ad \gg 1$, which is the case for many fast transforms.

Example 1: These factors are very general and include a large class of sparse matrices such as square dyadic butterfly factors [6], [7] and monarch matrices [8]:

- Square dyadic butterfly factors appear in sparse factorizations of many matrices associated to fast transforms, such as the Fast Fourier transform [1] and the Fast Walsh-Hadamard transform [9]. If the matrix has size $n \times n$ with $n = 2^L$, the factorization counts L butterfly factors. For the ℓ -th factor, $\ell \in \llbracket 1, L \rrbracket$, it holds $\pi_\ell = (2^{\ell-1}, 2, 2, 2^{L-\ell})$.
- Monarch factorizations have been used for weight matrices in transformers for vision or language tasks led

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to speed-ups of training and inference time. A monarch factorization is parameterized by two integers p, q and decomposes a matrix $\mathbf{A}$ of size $m \times n$ into 2 factors. Assuming that p, q divide m, n respectively, for the first factor it holds $\pi_1 = (1, p, q, m/p)$ and for the second factor it holds $\pi_2 = (q, m/p, n/q, 1)$.

III. MAIN RESULT

In this section, we will prove that if a matrix $\mathbf{K}$ admits a Kronecker-sparse factorization, then the matrix $\mathbf{K}' := \text{Re}(\mathbf{K})$ also admits a Kronecker-sparse factorization with a different chain. We will first show that $\text{Re}(\mathbf{K})$ can be expressed as a product of L matrices but without the Kronecker-sparse structure, and then we will apply multiple permutations in order to recover a Kronecker-sparse factorization.

A. Factorization of the real part

First, as being a Kronecker-sparse factor is only related to the support of the matrix. This means that if we have a single Kronecker-sparse factor, then its real and its imaginary part are also Kronecker-sparse factors with the same chain.

Now, we consider the case where we have a product of two Kronecker-sparse factors, $\mathbf{K}_1$ and $\mathbf{K}_2$. We want to show that the real part of the product $\mathbf{K}_1 \mathbf{K}_2$ is also a product of two Kronecker-sparse factors. With a simple linear algebra computation, we have

$$\begin{aligned} \text{Re}(\mathbf{K}_1 \mathbf{K}_2) &= \text{Re}(\mathbf{K}_1) \text{Re}(\mathbf{K}_2) - \text{Im}(\mathbf{K}_1) \text{Im}(\mathbf{K}_2) \\ &= (\text{Re}(\mathbf{K}_1) \quad \text{Im}(\mathbf{K}_1)) \cdot \begin{pmatrix} \text{Re}(\mathbf{K}_2) \\ -\text{Im}(\mathbf{K}_2) \end{pmatrix}, \end{aligned} \quad (1)$$

$$\begin{aligned} \text{Im}(\mathbf{K}_1 \mathbf{K}_2) &= \text{Re}(\mathbf{K}_1) \text{Im}(\mathbf{K}_2) + \text{Im}(\mathbf{K}_1) \text{Re}(\mathbf{K}_2) \\ &= (\text{Re}(\mathbf{K}_1) \quad \text{Im}(\mathbf{K}_1)) \cdot \begin{pmatrix} \text{Im}(\mathbf{K}_2) \\ \text{Re}(\mathbf{K}_2) \end{pmatrix}. \end{aligned} \quad (2)$$

Now, for $L = 3$, the same argument gives:

$$\text{Re}(\mathbf{K}_1 \mathbf{K}_2 \mathbf{K}_3) = (\text{Re}(\mathbf{K}_1 \mathbf{K}_2) \quad \text{Im}(\mathbf{K}_1 \mathbf{K}_2)) \cdot \begin{pmatrix} \text{Re}(\mathbf{K}_3) \\ -\text{Im}(\mathbf{K}_3) \end{pmatrix}.$$

Now, we have to express $(\text{Re}(\mathbf{K}_1 \mathbf{K}_2) \quad \text{Im}(\mathbf{K}_1 \mathbf{K}_2))$ as a product of two Kronecker-sparse factors. To do so, we use (1) and (2) to obtain

$$\begin{aligned} &(\text{Re}(\mathbf{K}_1 \mathbf{K}_2) \quad \text{Im}(\mathbf{K}_1 \mathbf{K}_2)) \\ &= (\text{Re}(\mathbf{K}_1) \quad \text{Im}(\mathbf{K}_1)) \cdot \begin{pmatrix} \text{Re}(\mathbf{K}_2) & \text{Im}(\mathbf{K}_2) \\ -\text{Im}(\mathbf{K}_2) & \text{Re}(\mathbf{K}_2) \end{pmatrix}. \end{aligned}$$

Therefore, the final factorization is

$$\begin{aligned} &\text{Re}(\mathbf{K}_1 \mathbf{K}_2 \mathbf{K}_3) \\ &= (\text{Re}(\mathbf{K}_1) \quad \text{Im}(\mathbf{K}_1)) \cdot \begin{pmatrix} \text{Re}(\mathbf{K}_2) & \text{Im}(\mathbf{K}_2) \\ -\text{Im}(\mathbf{K}_2) & \text{Re}(\mathbf{K}_2) \end{pmatrix} \cdot \begin{pmatrix} \text{Re}(\mathbf{K}_3) \\ -\text{Im}(\mathbf{K}_3) \end{pmatrix}. \end{aligned}$$

Recursively, we can show that

$$\begin{aligned} \text{Re}(\mathbf{K}_1 \cdots \mathbf{K}_L) &= (\text{Re}(\mathbf{K}_1) \quad \text{Im}(\mathbf{K}_1)) \cdot \begin{pmatrix} \text{Re}(\mathbf{K}_2) & \text{Im}(\mathbf{K}_2) \\ -\text{Im}(\mathbf{K}_2) & \text{Re}(\mathbf{K}_2) \end{pmatrix} \\ &\cdots \begin{pmatrix} \text{Re}(\mathbf{K}_{L-1}) & \text{Im}(\mathbf{K}_{L-1}) \\ -\text{Im}(\mathbf{K}_{L-1}) & \text{Re}(\mathbf{K}_{L-1}) \end{pmatrix} \cdot \begin{pmatrix} \text{Re}(\mathbf{K}_L) \\ -\text{Im}(\mathbf{K}_L) \end{pmatrix}. \end{aligned}$$

We have shown that if $\mathbf{K} := \mathbf{K}_1 \cdots \mathbf{K}_L$ with a chain $(\pi_1, \dots, \pi_L)$ for $\pi_\ell = (a_\ell, b_\ell, c_\ell, d_\ell)$, $\ell \in \llbracket 1, L \rrbracket$, admits a Kronecker-sparse factorization, then $\text{Re}(\mathbf{K}) = \tilde{\mathbf{K}}_1 \cdots \tilde{\mathbf{K}}_L$ where

$$\begin{aligned} \tilde{\mathbf{K}}_1 &:= (\text{Re}(\mathbf{K}_1) \quad \text{Im}(\mathbf{K}_1)) \\ \tilde{\mathbf{K}}_\ell &:= \begin{pmatrix} \text{Re}(\mathbf{K}_\ell) & \text{Im}(\mathbf{K}_\ell) \\ -\text{Im}(\mathbf{K}_\ell) & \text{Re}(\mathbf{K}_\ell) \end{pmatrix}, \quad 2 \leq \ell \leq L-1 \\ \tilde{\mathbf{K}}_L &:= \begin{pmatrix} \text{Re}(\mathbf{K}_L) \\ -\text{Im}(\mathbf{K}_L) \end{pmatrix}. \end{aligned} \quad (3)$$

However, these new factors $\tilde{\mathbf{K}}_\ell$, $\ell \in \llbracket 1, L \rrbracket$, are not Kronecker-sparse factors, because they do not have the right support. We have to find permutation matrices $\mathbf{P}_2, \dots, \mathbf{P}_L$ such that

$$\begin{aligned} \tilde{\mathbf{K}}_1 &:= \tilde{\mathbf{K}}_1 \mathbf{P}_2^\top \in \mathcal{K}_{(a_1, b_1, 2c_1, d_1)} \\ \tilde{\mathbf{K}}_\ell &:= \mathbf{P}_\ell \tilde{\mathbf{K}}_\ell \mathbf{P}_{\ell+1}^\top \in \mathcal{K}_{(a_\ell, 2b_\ell, 2c_\ell, d_\ell)}, \quad 2 \leq \ell \leq L-1 \\ \tilde{\mathbf{K}}_L &:= \mathbf{P}_L \tilde{\mathbf{K}}_L \in \mathcal{K}_{(a_L, 2b_L, c_L, d_L)}. \end{aligned} \quad (4)$$

For the choice of the support of the new factors, $\tilde{\mathbf{K}}_1$ and $\tilde{\mathbf{K}}_L$ have their number of rows (resp. columns) doubled, and the only way to act on it is to replace c_1 by $2c_1$ (resp. b_L by $2b_L$) in the new chain. For the factors $\tilde{\mathbf{K}}_\ell$, $2 \leq \ell \leq L-1$, in order to keep the number of non-zero elements, we need to replace b_ℓ and c_ℓ by $2b_\ell$ and $2c_\ell$, respectively.

Since $\mathbf{P}_\ell^\top \mathbf{P}_\ell = \mathbf{I}$, $\ell \in \llbracket 2, L \rrbracket$, we well obtain a Kronecker-sparse factorization of $\text{Re}(\mathbf{K}) = \tilde{\mathbf{K}}_1 \cdots \tilde{\mathbf{K}}_L$ with the new chain $\pi'_1, \dots, \pi'_L$ where

$$\pi'_\ell := \begin{cases} (a_1, b_1, 2c_1, d_1) & \text{if } \ell = 1 \\ (a_L, b_L, 2c_L, d_L) & \text{if } \ell = L \\ (a_\ell, 2b_\ell, 2c_\ell, d_\ell) & \text{otherwise.} \end{cases} \quad (5)$$

To satisfy the above conditions, the permutation matrices $\mathbf{P}_\ell$ have to satisfy the following conditions:

$$\mathbf{P}_2 \cdot \begin{pmatrix} \mathbf{S}_{\pi'_1}^\top \\ \mathbf{S}_{\pi'_1} \end{pmatrix} = \mathbf{S}_{(a_1, b_1, 2c_1, d_1)}^\top = \mathbf{S}_{(a_1, 2c_1, b_1, d_1)} \quad (6)$$

$$\mathbf{P}_\ell \cdot \begin{pmatrix} \mathbf{S}_{\pi'_\ell} & \mathbf{S}_{\pi'_\ell} \\ \mathbf{S}_{\pi'_\ell} & \mathbf{S}_{\pi'_\ell} \end{pmatrix} \cdot \mathbf{P}_{\ell+1}^\top = \mathbf{S}_{(a_\ell, 2b_\ell, 2c_\ell, d_\ell)}, \quad 2 \leq \ell \leq L-1 \quad (7)$$

$$\mathbf{P}_L \cdot \begin{pmatrix} \mathbf{S}_{\pi'_L} \\ \mathbf{S}_{\pi'_L} \end{pmatrix} = \mathbf{S}_{(a_L, 2b_L, c_L, d_L)}. \quad (8)$$

B. Finding the right permutation matrices

We continue by giving the explicit construction of the permutation matrices. We begin by defining the commutation matrix $\mathbf{C}_{m,n} \in \mathbb{R}^{mn \times mn}$ [10] where its permutation σ over the set $\llbracket 1, mn \rrbracket$ is defined as

$$\sigma(i + m(j-1)) = j + n(i-1), \quad i \in \llbracket 1, m \rrbracket, j \in \llbracket 1, n \rrbracket.$$

As this matrix is the commutation matrix, its action on left with the particular choice $m = 2$ is to permute the lines by grouping on top the lines of indices $i = 0$ and $i = m$, then the lines of indices $i = 1$ and $i = m+1$, and so on. Hence, we have the following lemma, which will be useful in the following.

Lemma 1: Given $\mathbf{A} \in \mathbb{C}^{m \times n}$, we have

$$\mathbf{C}_{2,m} \begin{pmatrix} \mathbf{A} \\ \mathbf{A} \end{pmatrix} = \mathbf{C}_{2,m} \cdot (\mathbf{1}_{2 \times 1} \otimes \mathbf{A}) = \mathbf{A} \otimes \mathbf{1}_{2 \times 1}.$$

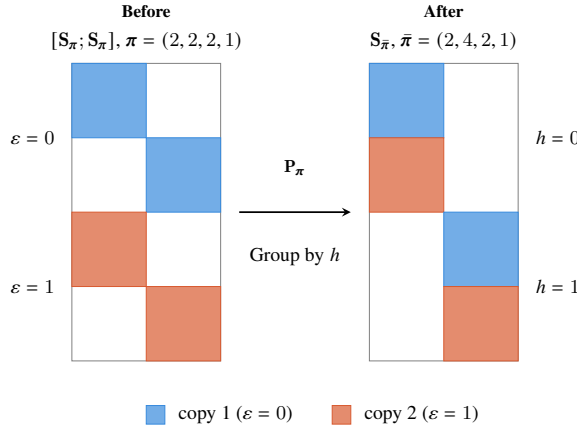


Fig. 1. Illustration of the action of the permutation matrix $\mathbf{P}_\pi$ on the concatenated Kronecker-sparse factors with chain $\pi = (a, b, c, d)$.

Proof: The proof is a direct consequence of the definition of the commutation matrix, the property of the Kronecker product, and the fact that $\mathbf{C}_{1,n} = \mathbf{I}_n$ for every $n \in \mathbb{N}$. ■

Then, given a chain $\pi = (a, b, c, d)$, we define the permutation matrix

$$\mathbf{P}_\pi := \mathbf{C}_{2,a} \otimes \mathbf{I}_{bd}. \quad (9)$$

This matrix has the same effect as the commutation matrix but on groups of bd consecutive lines instead of single lines. Its associated row-permutation, ρ_π , acts on rows of the matrix $\mathbf{1}_{2 \times 1} \otimes \mathbf{A}$, where $\mathbf{A} \in \mathbb{R}^{bd \times cd}$, over the set $\llbracket 1, 2abd \rrbracket$ as

$$\begin{aligned} \rho_\pi((\varepsilon a + h)bd + j) &= (2h + \varepsilon)bd + j, \\ \varepsilon &\in \{0, 1\}, h \in \llbracket 0, a - 1 \rrbracket, j \in \llbracket 0, bd - 1 \rrbracket. \end{aligned}$$

Fig. 1 illustrates the action of the permutation matrix $\mathbf{P}_\pi$ on the concatenated Kronecker-sparse factors with chain $\pi = (a, b, c, d)$.

Hence, applying this permutation matrix on the left to our concatenated Kronecker-sparse factors with chain π will group the lines in order to obtain a Kronecker-sparse factor with chain $\bar{\pi} := (a, 2b, c, d)$. This is formalized in the following lemma.

Lemma 2: Given $\pi = (a, b, c, d)$, the matrix $\mathbf{P}_\pi$ satisfies

$$\mathbf{P}_\pi \cdot \begin{pmatrix} \mathbf{S}_\pi \\ \mathbf{S}_\pi \end{pmatrix} = \mathbf{S}_{\bar{\pi}} \in \mathbb{R}^{2abd \times acd},$$

where $\bar{\pi} := (a, 2b, c, d)$.

Proof: With the fact that $\mathbf{S}_{(a,b,c,d)} = \mathbf{I}_a \otimes \mathbf{1}_{b \times c} \otimes \mathbf{I}_d$, the mixed-product property of the Kronecker product and Lemma 1, we have

$$\begin{aligned} \mathbf{P}_\pi \cdot \begin{pmatrix} \mathbf{S}_\pi \\ \mathbf{S}_\pi \end{pmatrix} &= (\mathbf{C}_{2,a} \otimes \mathbf{I}_{bd}) \cdot (\mathbf{1}_{2 \times 1} \otimes \mathbf{I}_a \otimes \mathbf{1}_{b \times c} \otimes \mathbf{I}_d) \\ &= (\mathbf{C}_{2,a} \cdot (\mathbf{1}_{2 \times 1} \otimes \mathbf{I}_a)) \otimes (\mathbf{I}_{bd}(\mathbf{1}_{b \times c} \otimes \mathbf{I}_d)) \\ &= (\mathbf{I}_a \otimes \mathbf{1}_{2 \times 1}) \otimes (\mathbf{1}_{b \times c} \otimes \mathbf{I}_d) \\ &= \mathbf{I}_a \otimes \underbrace{(\mathbf{1}_{2 \times 1} \otimes \mathbf{1}_{b \times c})}_{=\mathbf{1}_{2b \times c}} \otimes \mathbf{I}_d = \mathbf{S}_{\bar{\pi}}. \end{aligned}$$

With this lemma, we can give a new condition on the permutation matrices to satisfy (6), (7) and (8).

Lemma 3: For any sequence of chains $(\pi_1, \dots, \pi_L)$ with $\pi_\ell = (a_\ell, b_\ell, c_\ell, d_\ell)$, $\ell \in \llbracket 1, L \rrbracket$, (8) is satisfied with $\mathbf{P}_L := \mathbf{P}_{\pi_L}$. Furthermore, (6) and (7) are satisfied with $\mathbf{P}_\ell := \mathbf{P}_{\pi_\ell}$ if and only if, for every $\ell \in \llbracket 1, L - 1 \rrbracket$

$$\begin{aligned} &(\mathbf{C}_{2,a_{\ell+1}} \otimes \mathbf{I}_{b_{\ell+1}d_{\ell+1}}) \cdot \\ &\left(\mathbf{1}_{2 \times 1} \otimes \mathbf{I}_{a_\ell} \otimes \mathbf{1}_{c_\ell \times b'_\ell} \otimes \mathbf{I}_{d_\ell} \right) = \mathbf{I}_{a_\ell} \otimes \mathbf{1}_{2c_\ell \times b'_\ell} \otimes \mathbf{I}_{d_\ell}, \end{aligned} \quad (10)$$

where $b'_\ell = b_\ell$ if $\ell = 1$ and $b'_\ell = 2b_\ell$ otherwise.

Proof: First, as a consequence of Lemma 2, we have

$$\mathbf{P}_{\pi_\ell} \cdot \begin{pmatrix} \mathbf{S}_{\pi_\ell} \\ \mathbf{S}_{\pi_\ell} \end{pmatrix} = \mathbf{S}_{\bar{\pi}_\ell}, \quad 2 \leq \ell \leq L,$$

so, (8) is satisfied with $\mathbf{P}_L := \mathbf{P}_{\pi_L}$.

We also have that

$$\mathbf{P}_{\pi_\ell} \cdot \begin{pmatrix} \mathbf{S}_{\pi_\ell} & \mathbf{S}_{\pi_\ell} \\ \mathbf{S}_{\pi_\ell} & \mathbf{S}_{\pi_\ell} \end{pmatrix} = \begin{pmatrix} \mathbf{S}_{\bar{\pi}_\ell} & \mathbf{S}_{\bar{\pi}_\ell} \\ \mathbf{S}_{\bar{\pi}_\ell} & \mathbf{S}_{\bar{\pi}_\ell} \end{pmatrix}, \quad 2 \leq \ell \leq L.$$

If we denote $\bar{\pi} := (a, 2b, 2c, d)$ for a chain $\pi := (a, b, c, d)$, then (7) is equivalent to

$$(\mathbf{S}_{\bar{\pi}_\ell} \quad \mathbf{S}_{\bar{\pi}_\ell}) \cdot \mathbf{P}_{\pi_{\ell+1}}^\top = \mathbf{S}_{\bar{\pi}_\ell}, \quad 2 \leq \ell \leq L - 1,$$

which can be rewritten as

$$\mathbf{P}_{\pi_{\ell+1}} \cdot \begin{pmatrix} \mathbf{S}_{(a_\ell, c_\ell, 2b_\ell, d_\ell)} \\ \mathbf{S}_{(a_\ell, c_\ell, 2b_\ell, d_\ell)} \end{pmatrix} = \mathbf{S}_{(a_\ell, 2c_\ell, 2b_\ell, d_\ell)}, \quad 2 \leq \ell \leq L - 1.$$

Finally, (6) is equivalent to (for $\ell = 1$)

$$\mathbf{P}_{\pi_{\ell+1}} \cdot \begin{pmatrix} \mathbf{S}_{(a_\ell, c_\ell, b_\ell, d_\ell)} \\ \mathbf{S}_{(a_\ell, c_\ell, b_\ell, d_\ell)} \end{pmatrix} = \mathbf{S}_{(a_\ell, 2c_\ell, b_\ell, d_\ell)}.$$

This lemma transforms a condition on the permutation matrices into a condition on the chain of the Kronecker-sparse factors and allows us to characterize the existence of such a factorization. With our permutation matrices, we can give, in the following proposition, the three sufficient conditions on the chain of the Kronecker-sparse factors to ensure that the real part of the product of these factors also admits a Kronecker-sparse factorization with the new chain defined in (5). ■

Proposition 1: The Kronecker-sparse factorization of $\text{Re}(\mathbf{K}) = \tilde{\mathbf{K}}_1 \cdots \tilde{\mathbf{K}}_L$ exists with the new chain $(\pi'_1, \dots, \pi'_L)$ where π'_ℓ is defined in (5), if and only if, for every $\ell \in \llbracket 1, L - 1 \rrbracket$, the three following conditions hold:

- C1. $a_\ell c_\ell d_\ell = a_{\ell+1} b_{\ell+1} d_{\ell+1}$;
- C2. $\frac{a_\ell c_\ell}{a_{\ell+1}} = \frac{b_{\ell+1} d_{\ell+1}}{d_\ell} \in \mathbb{N}$;
- C3. $a_\ell | a_{\ell+1}$.

Proof: C1 is necessary and sufficient to ensure that the dimensions of the matrices are compatible for the matrix multiplication.

a) *Sufficiency of C2 and C3:* We denote by $r := \frac{a_\ell c_\ell}{a_{\ell+1}} = \frac{b_{\ell+1} d_{\ell+1}}{d_\ell}$, which is an integer by the second condition. Moreover, as $a_\ell | a_{\ell+1}$, there exists an integer p such that $p := \frac{a_{\ell+1}}{a_\ell}$, so $c_\ell = pr$. Hence, we can decompose the matrix $\mathbf{1}_{c_\ell \times b'_\ell} = \mathbf{1}_{p \times b'_\ell} \otimes \mathbf{1}_{r \times 1}$ then use the mixed-product property

of the Kronecker product, therefore the left-hand side of (10) can be rewritten as

$$\begin{aligned}
 & (\mathbf{C}_{2,a_{\ell+1}} \otimes \mathbf{I}_{d_{\ell+1}b_{\ell+1}}) \cdot \left(\mathbf{1}_{2 \times 1} \otimes \mathbf{I}_{a_{\ell}} \otimes \mathbf{1}_{c_{\ell} \times b'_{\ell}} \otimes \mathbf{I}_{d_{\ell}} \right) \\
 &= (\mathbf{C}_{2,a_{\ell+1}} \otimes \mathbf{I}_{d_{\ell+1}b_{\ell+1}}) \cdot \left(\mathbf{1}_{2 \times 1} \otimes \mathbf{I}_{a_{\ell}} \otimes \mathbf{1}_{p \times b'_{\ell}} \otimes \mathbf{1}_{r \times 1} \otimes \mathbf{I}_{d_{\ell}} \right) \\
 &= \mathbf{C}_{2,a_{\ell+1}} \cdot \left(\mathbf{1}_{2 \times 1} \otimes \mathbf{I}_{a_{\ell}} \otimes \mathbf{1}_{p \times b'_{\ell}} \right) \cdot (\mathbf{I}_{d_{\ell+1}b_{\ell+1}} \cdot (\mathbf{1}_{r \times 1} \otimes \mathbf{I}_{d_{\ell}})) \\
 &= \mathbf{I}_{a_{\ell}} \otimes \mathbf{1}_{2p \times b'_{\ell}} \otimes \mathbf{1}_{r \times 1} \otimes \mathbf{I}_{d_{\ell}} \\
 &= \mathbf{I}_{a_{\ell}} \otimes \mathbf{1}_{2c_{\ell} \times b'_{\ell}} \otimes \mathbf{I}_{d_{\ell}}.
 \end{aligned}$$

b) *Necessity of C2 and C3:* In the following, we denote by $\mathbf{X}_{\ell} := \mathbf{1}_{2 \times 1} \otimes \mathbf{I}_{a_{\ell}} \otimes \mathbf{1}_{c_{\ell} \times b'_{\ell}} \otimes \mathbf{I}_{d_{\ell}}$, $\mathbf{Y}_{\ell} := \mathbf{I}_{a_{\ell}} \otimes \mathbf{1}_{2c_{\ell} \times b'_{\ell}} \otimes \mathbf{I}_{d_{\ell}}$, $\ell \in \llbracket 1, L-1 \rrbracket$, therefore (10) is equivalent to $\mathbf{P}_{\pi_{\ell+1}} \cdot \mathbf{X}_{\ell} = \mathbf{Y}_{\ell}$.

We continue by indexing the rows of $\mathbf{X}_{\ell}$ by $(\varepsilon, i, k_1, k_2) \in \{0, 1\} \times \llbracket 0, a_{\ell} - 1 \rrbracket \times \llbracket 0, c_{\ell} - 1 \rrbracket \times \llbracket 0, d_{\ell} - 1 \rrbracket$ via $\varepsilon(a_{\ell}c_{\ell}d_{\ell}) + i(c_{\ell}d_{\ell}) + k_1d_{\ell} + k_2$, and the columns of $\mathbf{X}_{\ell}$ (equivalently of $\mathbf{Y}_{\ell}$ because they live in the same column space of size $a_{\ell}b'_{\ell}d_{\ell}$) by $(i', j, k'_2) \in \llbracket 0, a_{\ell} - 1 \rrbracket \times \llbracket 0, b'_{\ell} - 1 \rrbracket \times \llbracket 0, d_{\ell} - 1 \rrbracket$. Since $\mathbf{X}_{\ell} = \mathbf{1}_{2 \times 1} \otimes \mathbf{I}_{a_{\ell}} \otimes \mathbf{1}_{c_{\ell} \times b'_{\ell}} \otimes \mathbf{I}_{d_{\ell}}$

$$\mathbf{X}_{\ell}[(\varepsilon, i, k_1, k_2), (i', j, k'_2)] = \delta_{\varepsilon, i'} \delta_{k_2, k'_2}.$$

We now index the rows of $\mathbf{Y}_{\ell}$ by $(i, k'_1, k_2) \in \llbracket 0, a_{\ell} - 1 \rrbracket \times \llbracket 0, 2c_{\ell} - 1 \rrbracket \times \llbracket 0, d_{\ell} - 1 \rrbracket$ via $i(2c_{\ell}d_{\ell}) + k'_1d_{\ell} + k_2$. Since $\mathbf{Y}_{\ell} = \mathbf{I}_{a_{\ell}} \otimes \mathbf{1}_{2c_{\ell} \times b'_{\ell}} \otimes \mathbf{I}_{d_{\ell}}$, we have

$$\mathbf{Y}_{\ell}[(i, k'_1, k_2), (i', j, k'_2)] = \delta_{i, i'} \delta_{k_2, k'_2}.$$

In particular, for each fixed (i, k_2) , the set of rows of $\mathbf{X}_{\ell}$ with first/last coordinates (i, k_2) (there are $2c_{\ell}$ of them indexed by (ε, k_1)) and the set of rows of $\mathbf{Y}_{\ell}$ with first/last coordinates (i, k_2) (there are $2c_{\ell}$ of them indexed by k'_1) have identical entries. Hence, we obtain the equivalence

$$\begin{aligned}
 \mathbf{P}_{\pi_{\ell+1}} \mathbf{X}_{\ell} = \mathbf{Y}_{\ell} &\iff \rho_{\pi_{\ell+1}} \left(\{(\varepsilon, i, k_1, k_2)\}_{\varepsilon \in \{0,1\}, k_1 \in \llbracket 0, c_{\ell}-1 \rrbracket} \right) \\
 &= \{(i, k'_1, k_2)\}_{k'_1 \in \llbracket 0, 2c_{\ell}-1 \rrbracket}, \\
 \forall (i, k_2) &\in \llbracket 0, a_{\ell} - 1 \rrbracket \times \llbracket 0, d_{\ell} - 1 \rrbracket.
 \end{aligned} \tag{11}$$

With this equivalence, we can now prove the necessity of C2. We consider the row of $\mathbf{X}_{\ell}$ indexed by $(1, 0, 0, 0)$, i.e., the row is indexed by $a_{\ell}c_{\ell}d_{\ell} = a_{\ell+1}b_{\ell+1}d_{\ell+1}$ with C1, then we have

$$\rho_{\pi_{\ell+1}}(a_{\ell+1}b_{\ell+1}d_{\ell+1}) = (2 \cdot 0 + 1)b_{\ell+1}d_{\ell+1} = b_{\ell+1}d_{\ell+1}.$$

According to (11), there exists $k'_1 \in \llbracket 0, 2c_{\ell} - 1 \rrbracket$ such that $b_{\ell+1}d_{\ell+1} = k'_1d_{\ell}$, which means that $\frac{b_{\ell+1}d_{\ell+1}}{d_{\ell}} \in \mathbb{N}$. And with C1, we have $\frac{b_{\ell+1}d_{\ell+1}}{d_{\ell}} = \frac{a_{\ell}c_{\ell}}{a_{\ell+1}} \in \mathbb{N}$.

Finally, for the necessity of C3, we first assume that $a_{\ell} \geq 2$ (otherwise $a_{\ell} = 1$ and the condition is trivially satisfied), we consider the row of $\mathbf{X}_{\ell}$ indexed by $(0, 1, 0, 0)$, i.e., the row is indexed by $c_{\ell}d_{\ell}$. Then, we perform an Euclidean division of c_{ℓ} by $r := \frac{a_{\ell}c_{\ell}}{a_{\ell+1}} = \frac{b_{\ell+1}d_{\ell+1}}{d_{\ell}}$: $c_{\ell} = pr + q$ with $p, q \in \mathbb{N}$ and $q < r$. So, with C1, $c_{\ell}d_{\ell} = prd_{\ell} + qd_{\ell} = pb_{\ell+1}d_{\ell+1} + qd_{\ell}$ and $qd_{\ell} < rd_{\ell} = b_{\ell+1}d_{\ell+1}$. Therefore, we get

$$\rho_{\pi_{\ell+1}}(c_{\ell}d_{\ell}) = \rho_{\pi_{\ell+1}}(pb_{\ell+1}d_{\ell+1} + qd_{\ell}) = (2pr + q)d_{\ell}.$$

According to (11), there exists $k'_1 \in \llbracket 0, 2c_{\ell} - 1 \rrbracket$ such that $(2pr + q)d_{\ell} = 2c_{\ell}d_{\ell} + k'_1d_{\ell} = (2pr + 2q + k'_1)d_{\ell}$, which means

that $q \leq 0$, so $q = 0$. Therefore, $r|c_{\ell}$, and since $r = \frac{a_{\ell}c_{\ell}}{a_{\ell+1}}$, we have $a_{\ell}|a_{\ell+1}$. ■

Remark 1 (Numerical corroboration): As a sanity check of the proof above, we also verified the equivalence of Proposition 1 numerically: we exhaustively tested it on several curated configurations and on several hundred randomly generated configurations of $(a_{\ell}, b'_{\ell}, c_{\ell}, d_{\ell}, a_{\ell+1}, b_{\ell+1}, d_{\ell+1})$ satisfying C1, and found no counterexample to the equivalence (10) $\iff$ C2 $\wedge$ C3.

A direct consequence of this proposition is the following corollary showing that every chainable architecture¹ satisfies the three conditions of the proposition.

Corollary 1: If a matrix $\mathbf{K}$ admits a Kronecker-sparse factorization with a chainable architecture [3, Definition 4.12], then $\text{Re}(\mathbf{K})$ also admits a Kronecker-sparse factorization with the new chain $(\pi'_1, \dots, \pi'_L)$ where π'_ℓ is defined as in (5).

The chainability condition is easier to check than the three conditions of Proposition 1, which makes this corollary a more practical result. For example, we know that the square dyadic butterfly factors (and the monarch factors) is a chainable architecture, so we can directly conclude with this corollary that the DCT and the DST admit a Kronecker-sparse factorization.

Remark 2 (On the Discrete Hartley Transform): The same results hold for the imaginary part of the product of Kronecker-sparse factors, with the same conditions on the chain of the factors. Indeed, $\text{Im}(\mathbf{K}) = \tilde{\mathbf{K}}_1 \cdots \tilde{\mathbf{K}}_L$ with $\tilde{\mathbf{K}}_1, \dots, \tilde{\mathbf{K}}_{L-1}$ defined in (3) and $\tilde{\mathbf{K}}_L := \begin{pmatrix} \text{Im}(\mathbf{K}_L) \\ \text{Re}(\mathbf{K}_L) \end{pmatrix}$. Finally, the same permutation matrices $\mathbf{P}_2, \dots, \mathbf{P}_L$ defined in (9) can be used to obtain a Kronecker-sparse factorization of $\text{Im}(\mathbf{K})$ with the new chain $(\pi'_1, \dots, \pi'_L)$ defined in (5). Furthermore, for any $\alpha, \beta \in \mathbb{R}$, we have $\alpha \text{Re}(\mathbf{K}) + \beta \text{Im}(\mathbf{K}) = \tilde{\mathbf{K}}_1 \cdots \tilde{\mathbf{K}}_L$ with $\tilde{\mathbf{K}}_1, \dots, \tilde{\mathbf{K}}_{L-1}$ defined in (3) and $\tilde{\mathbf{K}}_L := \begin{pmatrix} \alpha \text{Re}(\mathbf{K}_L) + \beta \text{Im}(\mathbf{K}_L) \\ -\beta \text{Re}(\mathbf{K}_L) + \alpha \text{Im}(\mathbf{K}_L) \end{pmatrix}$. Since a sum of matrices with the same support keeps the same support, we can use the same permutation matrices $\mathbf{P}_2, \dots, \mathbf{P}_L$ defined in (9) to obtain a Kronecker-sparse factorization of $\alpha \text{Re}(\mathbf{K}) + \beta \text{Im}(\mathbf{K})$ with the new chain $(\pi'_1, \dots, \pi'_L)$ defined in (5). Especially, with $\alpha = 1$ and $\beta = -1$, we recover the decomposition of the Discrete Hartley Transform as a sum of the DCT and the DST, which admits a Kronecker-sparse factorization according to the results above.

IV. CONCLUSION

In this letter, we addressed the question of whether the real or the imaginary part of a product of structured-sparse matrices, called Kronecker-sparse factors, retains the same structure. We provided a necessary and sufficient condition on the factors of the product to ensure this property. As a direct consequence, we proved that the Discrete Cosine, Sine and Hartley Transforms all admit a Kronecker-sparse factorization, which is a new result in the literature. This result is of practical interest and has been implemented in the python library `lazylinop`² for the design of efficient algorithms for these transforms.

¹This condition ensures that the product of two Kronecker-sparse factors is also a Kronecker-sparse factor.

²<https://faustgrp.gitlabpages.inria.fr/lazylinop/>

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